

Software Requirement Specification (SRS)

Project Name

Ultimate Customizable Product Platform

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1 Introduction

1.1 Purpose

This document specifies the software requirements for the Ultimate Customizable Product Platform, a web-based system that allows customers to design and order personalized merchandise such as apparel, mugs, stickers, and home decor. The platform brings online product customization together with real-time design tools, enabling customers to build personalized products, preview them instantly, and see dynamic pricing update as they work. The purpose of this SRS is to define the functional and non-functional requirements needed to guide the design, development, and evaluation of the platform, with particular emphasis on the integrated customization workflow and the secure Open Delivery verification process that protects customers at the point of delivery.

1.2 Scope

The Ultimate Customizable Product Platform is a web-based application that provides the following functionalities:

- User Registration
- Login
- Product Browsing
- Product Customization
- Image Upload
- Text Editing
- Integrated Image & Sticker Search Engine
- Shopping Cart
- Payment
- Order Tracking
- Admin Panel
- Open Delivery

The platform aims to make personalized products accessible to everyday customers while giving them full confidence in the quality of what they receive.

1.3 Objectives

- Provide an easy, intuitive customization experience for non-technical users.
- Enable truly personalized products through flexible design tools.
- Deliver a better overall customer experience from design to delivery.
- Guarantee secure delivery verification through the Open Delivery protocol.
- Support a fast, streamlined ordering process from cart to checkout.
- Apply accurate, real-time dynamic pricing based on chosen customizations.

1.4 Intended Audience

- Customers — end users who design, order, and receive customized products.
- Administrators — staff who manage products, users, orders, and reports.
- Developers — the technical team responsible for building and maintaining the platform.
- Project Evaluators — academic reviewers assessing this project.
- Printing Vendors — partners who fulfil print-ready design files.
- Delivery Partners — personnel responsible for shipment and Open Delivery verification.

1.5 Definitions

Term	Description
Customer	A registered user who browses, customizes, and purchases products on the platform.
Admin	A platform operator who manages products, users, orders, and generates reports.
Customization	The set of design choices — text, images, colors, and layout — applied to a

Term	Description
	base product.
Sticker	A pre-loaded graphic, icon, or clipart asset that a customer can insert into a design.
Order	A finalized purchase consisting of one or more customized products, payment, and delivery details.
Open Delivery	A verification protocol that lets the customer inspect print quality before accepting an order.
Dynamic Pricing	The system's real-time recalculation of price as a customer adds or changes design elements.

2 Overall Description

2.1 Product Perspective

The Ultimate Customizable Product Platform is a self-contained web application built around a central Customization Studio, where customers design their products in real time. An integrated Image & Sticker Search Engine sits directly inside this studio, letting customers find and insert visual assets without breaking their design flow. Once a design is complete, the Shopping and Payment modules handle cart management and secure checkout, while the Admin module gives platform staff oversight of products, users, and orders. All platform data — user accounts, product catalogs, customization records, and order history — is maintained in a central database, and the Open Delivery module closes the loop by letting customers verify print quality at the moment of delivery.

2.2 Product Functions

At a high level, the system shall provide the following functions:

- Registration and Login
- Browse Products
- Customize Product
- Upload Images
- Search Images
- Search Stickers
- Add Text
- Resize, Rotate, and Layer Design Objects
- Live Preview
- Dynamic Price Calculation
- Add to Cart
- Checkout
- Order Tracking
- Admin Management

2.3 User Classes

Customer: Registers and logs in, browses the catalog, customizes products using the Design Studio, purchases orders, and tracks deliveries through Open Delivery.

Vendor: Receives the print-ready design files generated by the system and processes them into finished products.

Delivery Partner: Handles shipping and facilitates the Open Delivery inspection and OTP verification protocol at the customer's doorstep.

Admin: Manages the product catalog, user accounts, orders, and platform-wide reports.

2.4 Assumptions

- Users have a stable internet connection while designing and ordering.
- Users access the platform through a modern, up-to-date web browser.

- Customers upload images of sufficient quality for print reproduction.
- A third-party payment gateway is available and reachable at checkout.

2.5 Constraints

- The platform must support large-volume image storage for user uploads and sticker assets.
- The customization canvas must render designs quickly, without perceptible lag.
- All payment transactions must be processed securely.
- The user interface must remain responsive across desktop, tablet, and mobile devices.
- Uploaded images must be optimized to balance print quality against storage and load time.

3 Functional Requirements

FR1 — Registration

- Users shall be able to register using their name, email address, phone number, and a password.
- The system shall validate email format and enforce a minimum password strength.
- The system shall confirm successful registration before allowing the user to log in.

FR2 — Login

- The system shall allow users to log in using a registered email and password.
- The system shall authenticate users and redirect them to a role-specific dashboard (Customer or Admin).
- The system shall maintain a secure session and log out idle users automatically.

FR3 — Browse Products

- Users shall be able to browse the catalog of base products available for customization.
- Users shall be able to filter and search products by category, price, and material.
- Users shall be able to view full product details before starting a customization.

FR4 — Product Customization

- Base Product: Users shall select a base product and its available variants, including color and material.
- Text: Users shall be able to add custom text, choose fonts, adjust colors, and resize text elements.
- Image Upload: Users shall be able to upload personal images to include in a design.
- Layers: Users shall be able to resize, rotate, move, and reorder (layer) individual design elements.
- Preview: The system shall render a real-time preview of the customized product as changes are made.

FR5 — Integrated Image & Sticker Search Engine

The Image & Sticker Search Engine is the platform's core differentiating feature. It is embedded directly inside the Customization Page so that customers can find and use visual assets without ever leaving their design in progress.

- The search engine shall be accessible from within the customization page itself, not a separate screen.
- Users shall be able to search for images, icons, emojis, stickers, and SVG graphics from a single search bar.
- Search results shall appear instantly and support one-click insertion onto the design canvas.
- Users shall never be required to leave the customization page to find or add a design asset.
- Inserted objects shall support resizing, rotation, deletion, layering, and free movement.
- Inserted objects shall be combinable with the customer's own uploaded images within the same design.
- The dynamic price shall update automatically whenever a searched asset is added to or removed from the design.

FR6 — Shopping Cart

- Users shall be able to add one or more customized products to a shopping cart.
- The cart shall preserve the full design configuration of each item and allow quantity updates.

- The cart shall display a breakdown of base price, customization charges, and taxes.

FR7 — Payment

- The system shall support secure checkout through multiple payment options, including cards, UPI, and digital wallets.
- The system shall generate a unique Order ID upon successful payment.
- The system shall record payment details for order history and dispute handling.

FR8 — Order Management

- The system shall automatically generate a print-ready file package for each ordered design.
- Users shall be able to view current order status and complete order history.
- The system shall notify users at key order milestones, such as confirmation, printing, and shipping.

FR9 — Open Delivery

- Inspection: The delivery partner shall allow the customer to open and inspect the product before completing delivery.
- OTP: The system shall generate a secure Open Delivery OTP that the customer shares only after confirming the product is satisfactory.
- Accept: Once the OTP is confirmed, the order shall be marked as successfully delivered.
- Reject: If the product is damaged, incorrect, or below expected quality, the customer may reject it at the doorstep.
- Refund/Reprint: A rejected order shall automatically trigger a refund or reprint workflow, tracked through the Admin Panel.

FR10 — Admin Management

- Admins shall be able to add, update, and remove base products and variants in the catalog.
- Admins shall be able to view and, where necessary, suspend user accounts.
- Admins shall be able to view and manage all orders, including Open Delivery rejections and disputes.
- Admins shall be able to generate reports on sales, popular customizations, and overall platform activity.

4 Non-Functional Requirements

Performance

- The customization canvas shall respond to user actions such as dragging, resizing, and rotating within 100 milliseconds.
- Print-ready file generation shall run asynchronously in the background so it does not delay checkout.

Security

- All user-uploaded files shall be scanned before being processed or stored.
- Passwords shall be stored using secure, one-way hashing.
- Payment data shall be handled in compliance with recognized industry security standards.

Reliability

- The system shall maintain reliable operation with minimal unplanned downtime.
- Regular database backups shall be performed to prevent data loss.

Availability

- The platform shall remain accessible to users at all times, with any scheduled maintenance communicated in advance.

Scalability

- The architecture shall support growth in registered users, catalog size, and concurrent design sessions without redesign.

Maintainability

- The codebase shall follow a modular structure to simplify debugging, testing, and future feature additions.

Usability

- The interface shall be intuitive enough for a first-time user to complete a customization and place an order with minimal guidance.

Compatibility

- The platform shall be responsive and function correctly across modern browsers and across desktop, tablet, and mobile devices.

5 External Interface Requirements

User Interface

The application shall provide the following primary views:

- Home Page
- Login and Registration Pages
- Product Listing Page
- Customization Page
- Shopping Cart
- Checkout Page
- Order Tracking Page
- Admin Dashboard

Hardware Interface

The platform shall be accessible via Laptop, Desktop, Tablet, and Mobile devices.

Software Interface

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL, managed via phpMyAdmin
- Server: XAMPP
- IDE: Visual Studio Code
- Version Control: Git Bash
- Repository: GitHub

6 Database Design

The platform's data is organized around the following core tables. Each table is described conceptually below; detailed schema design is left to the implementation phase.

Table	Description
users	Stores registered customer and admin account details, including authentication credentials and contact information.
products	Stores the catalog of base products available for customization, including category, pricing, and available variants.
customizations	Stores each design configuration created by a customer, including selected text, colors, uploaded/searched assets, and layout.
stickers	Stores metadata for sticker, icon, and clipart assets surfaced through the integrated search engine.
images	Stores metadata and storage references for images uploaded by customers for use in their designs.
orders	Stores order details, including the linked customization, pricing, status, and Open Delivery verification data.

Table	Description
payments	Stores transaction records associated with each order, including payment method and status.
admins	Stores administrator account details and the access privileges assigned to platform staff.

7 Use Case Summary

Use Case	Actor(s)	Description
Register	Customer	A new user creates an account to access the platform.
Login	Customer, Admin	A registered user authenticates to reach their role-specific dashboard.
Browse Products	Customer	The customer views and filters the base product catalog.
Customize Product	Customer	The customer designs a product using the Customization Studio.
Search Images	Customer	The customer searches for images from within the customization page.
Search Stickers	Customer	The customer searches for stickers, icons, or clipart within the customization page.
Upload Images	Customer	The customer uploads a personal image for use in a design.
Checkout	Customer	The customer reviews the cart and proceeds toward payment.
Payment	Customer	The customer completes payment through a supported gateway.
Open Delivery	Customer, Delivery Partner	The customer inspects the product and accepts or rejects it at delivery using OTP verification.
Manage Orders	Admin	The admin views, updates, and resolves order and delivery-related issues.
Manage Catalog & Reports	Admin	The admin manages the product catalog and user accounts, and generates platform reports.

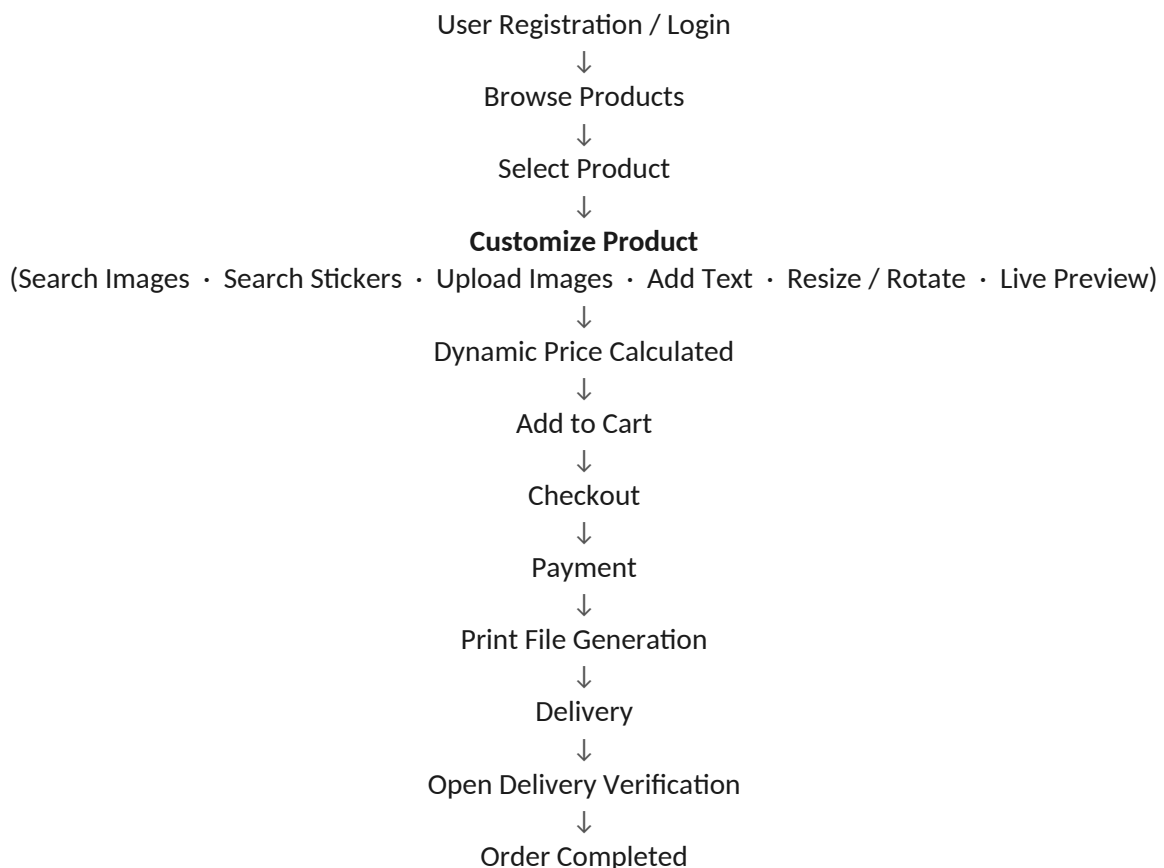
8 System Architecture & Technical Stack

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Delivers the responsive user interface, including the interactive Customization Studio.
Backend	PHP	Handles business logic, search integration, dynamic pricing, and order processing.
Database	MySQL (via phpMyAdmin)	Stores structured platform data, including users, products, orders, and customization records.
Server	XAMPP	Provides the local Apache, MySQL, and PHP

Layer	Technology	Purpose
		environment for development and testing.
IDE	Visual Studio Code	Primary environment for writing, debugging, and organizing project code.
Version Control	Git Bash	Enables command-line tracking of code changes across the development team.
Repository	GitHub	Hosts the project source code and supports collaborative development.

9 Data Flow

The diagram below traces a single customer journey from registration through to a completed, verified delivery.



The Image & Sticker Search Engine is fully integrated into the Customization Page: searching, inserting, and adjusting design assets all happen inline, without navigating away from the design in progress.

10 Conclusion

The Ultimate Customizable Product Platform aims to make personalized, print-on-demand products simple, transparent, and trustworthy to order online. By combining an intuitive Customization Studio with an integrated Image & Sticker Search Engine, the platform lets customers build a finished design in one continuous workflow, with dynamic pricing updating at every step. The Open Delivery protocol extends that trust to the final and most critical moment of the transaction, letting customers verify print quality before accepting an order. Built on a straightforward and well-understood technology stack, the platform is positioned to be extended over time with additional product categories, design tools, and delivery partnerships as it grows.